

Surprisal Modulates Dual-Task Performance in Older Adults: Pupillometry Shows Age-Related Trade-Offs in Task Performance and Time-Course of Language Processing

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Even though older adults are known to have difficulty at language processing when a secondary task has to be performed simultaneously, few studies have addressed how older adults process language in dual-task demands when linguistic load is systematically varied. Here, we manipulated surprisal, an information theoretic measure that quantifies the amount of new information conveyed by a word, to investigate how linguistic load affects younger and older adults during early and late stages of sentence processing under conditions when attention is split between two tasks. In high-surprisal sentences, target words were implausible and mismatched with semantic expectancies based on context, thereby causing integration difficulty. Participants performed semantic meaningfulness judgments on sentences that were presented in isolation (single task) or while performing a secondary tracking task (dual task). Cognitive load was measured by means of pupillometry. Mixed-effects models were fit to the data, showing the following: (a) During the dual task, younger but not older adults demonstrated early sensitivity to surprisal (higher levels of cognitive load, indexed by pupil size) as sentences were heard online; (b) Older adults showed no immediate reaction to surprisal, but a delayed response, where their meaningfulness judgments to high-surprisal words remained stable in accuracy, while secondary tracking performance declined. Findings are discussed in relation to age-related trade-offs in dual tasking and differences in the allocation of attentional resources during language processing. Collectively, our data show that higher linguistic load leads to task trade-offs in older adults and differently affects the time course of online language processing in aging.

Keywords: aging, dual tasking, pupillometry, sentence comprehension, surprisal

Even though language comprehension often seems like an effortless and very fast process, it is uncontroversial that the processing of new incoming information and its integration with earlier context can give rise to measurable integration difficulty that is resource-consuming (Baddeley, 1996; Clifton & Frazier, 1989; Gibson, 1991). One fundamental idea for measuring language-related processing difficulty is to quantify the amount of information that the linguistic signal conveys in a specific period of time. Early work on readability has put

forward the notion of “propositional density” (Kintsch & Keenan, 1973; Kintsch & Van Dijk, 1978), which quantifies the number of conceptual propositions or “idea units” per utterance. However, this notion does not take into account the semantic relationship between idea units—the more overlap two units share in lexical-semantic features, the easier it will be to integrate them. More recently, the information-theoretic measure of “surprisal” (Hale, 2001; Levy, 2008) has been proposed as a measure for quantifying the processing difficulty a comprehender incurs from a word, given its context.

Surprisal is defined as the negative log probability of a word appearing in a particular context ($-\log P(\text{word}|\text{context})$), and thus quantifies how predictable a word is, given its context (estimated by a word’s cloze probability). A word that is contextually unexpected has high surprisal and is predicted to cause greater difficulty than a word that can be anticipated based on context and has low surprisal. From the language production side, the distribution of surprisal is often referred to as “information density” (Jaeger, 2010; Levy & Jaeger, 2007), accounting for the experimental finding that speakers optimize the structure of their utterances by distributing information uniformly across the linguistic signal (thereby avoiding peaks or troughs in surprisal for the comprehender). In this paper, we use the conceptually broader phrase “informational density” as an umbrella term to refer to language structures that have higher levels of linguistic density. Under this

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notion, surprisal is a specific operationalization of informational density.

Indeed, surprisal has been shown to index processing difficulty and cognitive load in a range of psycholinguistic measures. For example, surprisal correlates with word reading times in eye-tracking experiments and with the N400 in electrophysiological studies (Boston, Hale, Kliegl, Patil, & Vasishth, 2008; Boston, Hale, Vasishth, & Kliegl, 2011; Demberg & Keller, 2008; Frank, Otten, Galli, & Vigliocco, 2013; Smith & Levy, 2013). Most relevant for the present study is its correlation with pupillometry (Demberg, Kiagia, & Sayeed, 2013; Demberg & Sayeed, 2016), a technique that is used to index cognitive load during mental operations by measuring stimulus-evoked dilations of the pupil (e.g., Beatty, 1982; Beatty & Lucero-Wagoner, 2000).

Given that words with high surprisal induce greater processing difficulty than more predictable words, the present study addresses two central questions: (a) How are high surprisal words comprehended in a dual-task situation when attentional resources to quickly resolve processing difficulties are less available because attention is already split between two tasks, and (b) how do the joint demands of high surprisal and a dual-task situation impact older adults, a population that is known to experience disproportional impairments in dual-tasking (Verhaeghen & Cerella, 2002; Verhaeghen, Steitz, Sliwinski, & Cerella, 2003). Based on prior literature addressing the effects of surprisal irrespective of aging (Hale, 2001; Levy, 2008), our expectation was that older adults should show increased processing difficulty (or even comprehension failure) for high-surprisal utterances during dual-task demands, because of the joint effects of processing difficulty associated with high-surprisal sentences on the one hand, and age-related limitations in dual-task ability on the other hand.

Age Impairments in Processing Informationally Dense Language

It is a well-documented finding in the psycholinguistic literature that healthy aging older adults show greater difficulty at comprehending language that is high in informational density. For example, older adults often have difficulty at processing and remembering sentences that are high in propositional density (e.g., Stine & Hindman, 1994; Stine & Wingfield, 1988), or with words or sentences that are encountered with no supporting discourse context and that are low in cloze probability (Madden, 1988; Steen-Baker et al., 2017; Wingfield, Poon, Lombardi, & Lowe, 1985).

Interestingly, informational density appears to differently impact the time course with which younger and older adults deploy attentional resources during language processing. For example, when reading longer text passages for immediate recall, older adults spend more time reading first-position sections, possibly because they put greater priority on creating a *gestalt* model of the text and its content first, and then use this information to facilitate comprehension of subsequent information (Stine-Morrow, Loveless, & Soederberg, 1996). Converging evidence from sentence processing suggests that, whereas younger adults register and resolve integration difficulties very early in the speech stream (immediately when the sentence is processed online), older adults tend to delay this process and then engage in subsequent repair during later processing stages (Shake & Stine-Morrow, 2011; Stites, Federmeier, & Stine-Morrow, 2013; for a model of atten-

tional allocation during language processing in old age, see Stine-Morrow, Soederberg Miller, Gagne, & Hertzog, 2008). Other studies have found qualitative changes in the way older adults use sentence context to predict upcoming information, suggesting that older adults rely more on context when the speech signal is visually or phonologically degraded (Madden, 1988; Wingfield et al., 1985), that older adults only predict the most highly expected exemplars cued by context (e.g., DeDe, 2014; Federmeier, McLennan, De Ochoa, & Kutas, 2002), or that older adults do not use context predictively at all (Wlotko, Federmeier, & Kutas, 2012; Wlotko, Lee, & Federmeier, 2010).

Age-Related Impairments in Dual-Tasking

In contrast to younger adults, older adults are notorious for their greater difficulty at performing two tasks at the same time (Verhaeghen & Cerella, 2002), including dual-task situations when one of the tasks requires processing of language (for language production tasks, see Becic et al., 2010; Kemper, Herman, & Lian, 2003; Kemper, Schmalzried, Herman, Leedahl, & Mohankumar, 2009; Kemper, Schmalzried, Hoffman, & Herman, 2010; Kemper et al., 2014; for comprehension tasks, see Tun, O'Kane, & Wingfield, 2002; Tun, Wingfield, & Stine, 1991). A common finding from these studies is that older adults adapt to dual-task situations differently than younger adults, by showing qualitatively different patterns in task prioritizing and task neglect (Becic et al., 2010; Kemper et al., 2009, 2010, 2014). For example, in a dual-task study that involved a story-retelling and a concurrent car driving task, older adults adapted to the dual-task demands by allocating all attentional resources to driving while neglecting the language task. Other studies have shown that older adults' speech production declines in dual-task situations: In spontaneous speech, older adults often sacrifice speech rate, coherence, grammatical complexity, and general fluency to maintain task performance (Kemper et al., 2003, 2009, 2010). One hypothesis to account for age-related trade-offs in dual tasking posits that older adults prioritize whatever task has a subjectively higher survival value to them. For example, when asked to perform a track-walking task and simultaneous memorizing of word lists, older adults prioritize walking over memory performance presumably because falls pose a greater health risk for older as opposed to younger adults (Lindenberger, Marsiske, & Baltes, 2000).

Dual-Task Studies on Processing of Informational Density in Younger and Older Adults

Only few studies have previously addressed how the joint effects of high informational density on the one hand, and dual-task demands on the other hand, play together to affect language comprehension in older adults. This lack of studies is somewhat surprising when we consider that we are faced with an increasing wealth of technological applications that allow us to perform multiple tasks simultaneously in our daily lives (e.g., talking to an intelligent artificial agent while driving, or talking on the phone while walking), and that older adults often complain about their difficulties at splitting attention in language comprehension (e.g., in listening to a conversation while ignoring others). What further complicates things is that the few dual-task studies that have actually investigated age-related changes in comprehension of

high informational density obtained somewhat conflicting findings. Drawing straightforward conclusions from these studies therefore remains impossible.

On the one hand, a study by Tun, O’Kane, and Wingfield (2002) showed that older adults demonstrated lower levels of recall for lexically scrambled sentences (i.e., sentences devoid in meaning or contextual support) when a secondary task required them to ignore meaningful sentences (as opposed to meaningless word strings). Given the greater dual-task effects in older adults specifically for the high-density meaningful sentences (as opposed to low-density, nonmeaningful words strings), these findings seem to provide some preliminary evidence for the notion that informationally dense language adds additional processing costs for older (and not so much younger) adults during the simultaneous execution of a secondary task.

On the other hand, contrasting findings emerged in two other studies (Kemper et al., 2014; Tun et al., 1991). Tun et al. (1991), for example, assessed age differences in a primary language task that required gist-wise recall of high- versus low-density text passages (measured as the number of ideas or propositions) and a secondary button-press reaction time (RT) task. According to the results, recall accuracy decreased as propositional density increased, an effect that was not modulated by age. In other words, older adults in that study showed no additional costs for recalling text passages that were high in informational density. Similar findings emerged in a study by Kemper et al. (2014), who investigated younger and older adults’ reading comprehension for text passages and sentences that varied in propositional density, grammatical complexity, and mean length of utterance (MLU) during execution of a simultaneous tracking task. Even though reading comprehension in that study declined as informational density increased in both age groups, older adults were not disproportionately affected by propositional density or grammatical complexity, neither in reading comprehension nor in their simultaneous tracking.

Unfortunately, none of these prior studies allow for detailed conclusions as to the processing difficulty that comprehenders encounter for single words in the speech stream, because they

relied upon measures of informational density that are primarily suitable to investigate language processing at a text and passage level (e.g., propositional content, grammatical complexity, MLU).

The Present Study

Here, we used surprisal (which quantifies the amount of information that each word conveys relative to the prior context) as a measure of informational density to investigate how first-pass processing and late-stage integration of words high in informational density is accomplished by groups of younger and older adults during the simultaneous execution of a secondary task. We used an auditory sentence comprehension and secondary tracking task (operationalized as a continuous driving task; see below). The tracking task allowed us to measure changes in steering performance as sentences unfolded in time. To constantly monitor processing difficulty during sentence processing, we assessed participants’ pupil size by means of an eye-tracker (Beatty, 1982; Beatty & Lucero-Wagoner, 2000). The pupillary response in both younger and older adults has been shown to be sensitive to increased cognitive (Van Gerven, Paas, Van Merriënboer, & Schmidt, 2004) and specifically linguistic load, for example in conditions when speech is masked by noise (Zekveld, Kramer, & Festen, 2010, 2011), or during comprehension of syntactically complex versus simple sentences (Just & Carpenter, 1993), notably during comprehension of syntactically complex sentences in older adults (Piquado, Isaacowitz, & Wingfield, 2010).

In the present study, we manipulated surprisal by presenting sentences that either fulfilled or violated semantic predictions based on prior context (low-surprisal: *Since Petra didn’t have anything to wear for the barbeque, she bought a [dress]_{critical word} in a nearby shop*; high-surprisal: *Since Petra didn’t have anything to drink for the barbeque, she bought a [dress]_{critical word} in a nearby shop*; see Table 1, for a schematic display). Note that surprisal was manipulated by the context words *wear* versus *drink*, so that the critical words in low- and high-surprisal sentences was always identical. High-surprisal sentences were thought to induce a strong cognitive conflict since the critical word in the second

Table 1
Experimental Items and Their English Translations

Surprisal	Item
Low surprisal	Weil Petra für den Grillabend nichts zum Anziehen hatte, kaufte sie sich ein Kleid in einem nahegelegenen Geschäft. <i>Since Petra didn’t have anything to wear for the barbeque, she bought a dress in a nearby shop.</i> Als Karl arbeitsuchend war, schickte er ganz viele Bewerbungen an ehemalige Kollegen. <i>When Karl was looking for work, he sent out a lot of applications to former colleagues.</i>
Low surprisal	Weil Petra für den Grillabend nichts zum Trinken hatte, kaufte sie sich ein Bier in einem nahegelegenen Geschäft. <i>Since Petra didn’t have anything to drink for the barbeque, she bought a beer in a nearby shop.</i> Als Karl verweist war, schickte er ganz viele Postkarten an ehemalige Kollegen. <i>When Karl was traveling, he sent out a lot of post cards to former colleagues.</i>
High surprisal	Weil Petra für den Grillabend nichts zum Trinken hatte, kaufte sie sich ein Kleid in einem nahegelegenen Geschäft. <i>Since Petra didn’t have anything to drink for the barbeque, she bought a dress in a nearby shop.</i> Als Karl arbeitsuchend war, schickte er ganz viele Postkarten an ehemalige Kollegen. <i>When Karl was looking for work, he sent out a lot of post cards to former colleagues.</i>
High surprisal	Weil Petra für den Grillabend nichts zum Anziehen hatte, kaufte sie sich ein Bier in einem nahegelegenen Geschäft. <i>Since Petra didn’t have anything to wear for the barbeque, she bought a beer in a nearby shop.</i> Als Karl verweist war, schickte er ganz viele Bewerbungen an ehemalige Kollegen. <i>When Karl was traveling, he sent out a lot of applications to former colleagues.</i>

Note. Critical words are printed in bold.

clause (*dress*) violated semantic context provided by the first clause (*drink*).

Our predictions for the results of this study were twofold. First, in line with prior literature, we predicted that dual-task costs should be more pronounced for older than younger adults, for example in the form of global performance declines or task trade-offs between language comprehension and tracking, or between earlier and later stages of sentence processing, in hindsight of prior studies demonstrating that older adults allocate attentional resources during language processing differently than younger adults (Shake & Stine-Morrow, 2011; Stine-Morrow et al., 2008; Stine-Morrow, Miller, & Hertzog, 2006; Stites et al., 2013). Specifically in relation to surprisal, we predicted that processing of high-surprisal sentences should be more effortful for older than younger adults. This effect should be particularly pronounced during the dual-task situation, because of the cumulative effects of age-related limitations in dual tasking on the one hand, and a surprisal-related increase in integration difficulty on the other hand.

Method

Participants

Thirty-two older adults participated for compensation at a rate of €10 per hour. The control group consisted of 32 younger adults, mostly students at Saarland University. Older adults were recruited through the UdS Psychology department participant pool, which regularly screens subjects for dementia. All older adults were neurologically intact and had no cognitive impairments. Younger adults were recruited through flyers on campus. Two older adults were excluded from further analysis because of below-chance performance in the AX-CPT task. The final sample included 32 younger adults (mean age = 25 years, age range = 19–34 years) and 30 older adults (mean age = 72 years, age range = 65–78 years). Based on self-report, all participants were native speakers of German, had normal or corrected-to-normal vision, and had no history of speech, hearing, language, and/or psychiatric disorders. The study was approved by the local ethics committee. All study procedures were in line with the Helsinki declaration on human subject testing. Informed written consent was obtained from all participants.

Participants were tested on a battery of cognitive tests, including the Digit Symbol Substitution task to assess processing speed (Wechsler, 1955), the Counting Span task to assess working memory (for review, see Conway et al., 2005), as well as a modified version of the AX-CPT task to assess context monitoring and updating (task adapted from Schmitt, Ferdinand, & Kray, 2014). Digit Symbol data for one older adult were lost because of technical malfunction. Sample characteristics as well as means and standard deviations for the cognitive tests are presented in Table 2.

Younger adults outperformed older adults in all measures of the cognitive test battery. Reaction times in the Digit Symbol task were significantly higher for older compared with younger adults, $t(52.54) = 8.97, p < .001$. There was also a significant main effect of age in the total number of correctly remembered trials for the Counting Span task, $t(57.14) = -2.89, p < .01$, with older adults remembering fewer trials correctly than younger adults (19 vs. 22, of 28 in total). In the AX-CPT task, older adults had longer RTs in both context-dependent, $t(59.23) = 4.19, p < .001$, and context-

Table 2

Sample Descriptives and Neuropsychological Test Performance in Younger and Older Adults

Measure	Younger adults	Older adults
<i>N</i>	32	30
Mean age	25 (3)	72 (4)
Gender (m f)	7 25	14 16
Handedness (l r)	6 26	0 30
Dominant eye (l r)	13 19	10 20
Neuropsychological tests		
Digit Symbol RT (in ms)	1148 (234)	1775 (304)
Counting Span (total correct of 28)	22 (4)	19 (5)
AX-CPT		
RT context-dependent (in ms)	760 (270)	1055 (284)
RT context-independent (in ms)	546 (136)	816 (237)
Accuracy context-dependent	.93 (.09)	.85 (.11)
Accuracy context-independent	.97 (.09)	.94 (.10)
RT cost (in ms)	214 (153)	239 (191)
Accuracy cost	.04 (.10)	.09 (.10)

Note. RT = reaction time. Standard deviations are shown in parentheses.

independent trials, $t(45.60) = -5.45, p < .001$, whereas for accuracy, the main effect of age was only significant for context-dependent trials, $t(52.34) = -2.82, p < .01$ (accuracy in context-independent trials: $t(58.04) = 1.32, p > .1$). However, there were no significant effects of age in either the RT cost score (difference in RT between context-dependent and context-independent trials, $t(53.30) = 0.65, p > .05$, or the accuracy cost score (difference in accuracy between context-independent and context-dependent trials), $t(55.95) = -1.45, p > .05$.

Materials

In the sentence comprehension task, participants performed semantic meaningfulness judgments for 188 auditorily presented sentences (47 items in four conditions). Half of the sentences were presented in a high-surprisal condition, the other half were presented in a low-surprisal condition. Each sentence consisted of two clauses, with the verb of the first clause providing a semantic context, and the noun of the second clause either matching or violating this semantic context (see Table 1). Each low- and high-surprisal item was also presented in a swapped condition. High-surprisal sentences were thought to induce greater integration difficulties than low-surprisal sentences, because of the semantic mismatch between prior context and critical word.

Items were classified as low- versus high-surprisal based on cloze probability ratings from two groups of younger ($n = 20$) and older adults ($n = 18$) who did not participate in the main experiment. Cloze probability was defined as the percentage of times a sentence context was completed with the critical word. All sentence contexts were created in a way so that they strongly biased a particular response ("high constraint"; see Federmeier, Wlotko, De Ochoa-Dewald, & Kutas, 2007; Wlotko et al., 2012). High-constraint context fragments paired with unexpected target words have previously been shown to elicit a reliable physiological (Wlotko et al., 2012) and also behavioral (Stine & Wingfield, 1994) response in both younger and older adults.

Based on the cloze ratings, low-surprisal items were constructed by choosing the most highly expected target word among the given

responses (the most frequent completion provided by the cloze ratings). In contrast, high-surprisal sentences were constructed by completing the sentence context with semantically incongruent words that violated prior context and were never indicated as possible targets in the cloze ratings. Thus, high-surprisal items in the current study had a very low cloze probability and also a strong competitor (i.e., the more likely but not actually mentioned completion of the sentence). Note that having a strong competitor is not part of the typical definition of surprisal (e.g., Hale, 2001).

On average, cloze probability ratings for low-surprisal sentences were higher in younger ($M = .86$, $SD = .13$) than in older adults ($M = .79$, $SD = .25$), a statistically significant difference, $t(140.88) = 2.60$, $p < .05$. However, post hoc inspection showed that this difference was largely driven by three items for which older adults' responses differed from those of younger adults lexically rather than semantically (for converging evidence, see Federmeier et al., 2002), for example, for a sentence which could be completed in "chimney" or "smokestack" (in German, "Kamin" vs. "Schornstein"), younger adults tended to use "smokestack" whereas older adults tended to use "chimney," or in a sentence which could be completed in "assessment" or "exam" (in German, "Prüfung" vs. "Klausur"), younger adults tended to use "Prüfung," whereas older adults tended to use "Klausur." Once these three items were excluded from the item pool, the group effect in cloze ratings was no longer significant, $t(144.30) = 1.66$, $p = .10$. To avoid an unbalanced design (which would have complicated fitting of models with maximal random slope structure), all items were left in the item pool for the analyses reported below.

To minimize prosodic differences among experimental items, the 188 sentences were synthesized using the MARY text-to-speech system (Schröder & Trouvain, 2003). Pauses were manipulated so that the duration of the critical word was always identical. To avoid stereotyped responses, we also presented 72 filler items, half in a high-surprisal condition (involving syntactic violations), the other half in a low-surprisal condition (no syntactic violation). All experimental items were counterbalanced over four experimental lists (corresponding to two single-language and two dual-language blocks) using a Latin square design, with surprisal and item as blocking factors. Thus, the number of low-surprisal and high-surprisal sentences participants encountered in one block and across the experiment was equal, and each subject encountered an item in only one of its four experimental conditions.

Design and Procedure

Participants came to our lab for one session that lasted approximately two hours. Upon filling in a demographic questionnaire, subjects completed the main experimental task that employed a dual-task paradigm to investigate age differences in processing high-surprisal sentences. In the dual-task condition, participants performed a sentence comprehension and a simultaneous tracking task (which was operationalized as a continuous driving task). In the single-task condition, participants performed the language task in isolation. Subjects listened to the experimental items over speakers, and were asked to judge whether the sentence they had just heard was meaningful by verbally answering "Yes" or "No," while the researcher recorded their responses. Overall, the experiment consisted of six experimental blocks: two dual-task blocks, two language-only blocks, and two tracking-only blocks, se-

quenced as driving single, language single, language dual, language dual, driving single, language single. Throughout the experiment, participants' pupil sizes were recorded by means of an eye-tracker.

To assess tracking performance, we used the Continuous Tracking and Reaction Task (ConTRe Task; Mahr, Feld, Moniri, & Math, 2012), a highly controlled tracking task that allows for fine-grained measurement of steering performance. Participants were seated in front of a steering wheel and saw a 3D road on a wrap-around screen, with two vertical color bars moving laterally on the road at a continuous speed (see Figure 1). They were instructed that they could control one of the bars (the blue [dark gray] one) by turning the steering wheel, whereas the other bar (the yellow [light gray] one) was continuously moving (controlled by the computer). The task was to continuously track the yellow bar with the blue one, to keep the distance between the two bars minimal at all times. Because the output variable (steering deviation; in meters) was encoded in positive and negative values (indicating left- and right-sided steering deviations), we squared its values to obtain a final measure. Thus, lower values in steering deviations in this task indicate better tracking performance. The cognitive test measures (Counting Span, Digit Symbol, and AX-CPT tasks) were administered after completion of the main experiment.

Apparatus

Throughout the experimental task, participants were seated behind the steering wheel, with the driving projection screen surrounding them and the eye-tracker installed at the top of the dashboard. We used OpenDS 3.0 (opens.de) to provide the driving environment, which was projected on three large panels positioned in a 180° angle around the car. The driving simulator, as well as the language comprehension task, were controlled from a separate computer using Experiment Builder (sr-research.com). Auditory items for the language comprehension task were pre-



Figure 1. The Continuous Tracking and Reaction Task (ConTRe) tracking task (Mahr et al., 2012). Participants are instructed to track the yellow (light gray) bar (which is continuously moving, controlled by the computer) with the blue (dark gray) bar by turning the steering wheel. Lumination is kept constant throughout the task. See the online article for the color version of this figure.

sented through Creative Gigaworks T20 speakers. Eye-tracking data were collected using the remote mode (no chin rest) of the Eye-link 1000 Plus system (SR-Research, Ontario, Canada), sampled at a rate of 250 Hz. Eye-tracking data were collected for both eyes. Before each experimental block, eye movements were calibrated using a nine-point grid. Participants were instructed to fixate on the blue bar (which remained fixed on the screen during single-task blocks), and to move their head as little as possible. The cognitive test battery was controlled by E-Prime and presented on a Dell Ultra Sharp U2515H computer screen, using a screen resolution of 2560×1440 pixels. The experimental session took place in a completely dark room, shielded from daylight. The wrap-around screen was the only source of luminance that participants directly faced, and its luminance remained constant throughout the experimental session (note that the mountain-road display and the vertical bars shown in Figure 1 remained on the screen even when participants were performing the single-language task with no tracking).

Results

Our dependent variables included steering deviations (squared) in the tracking task, response accuracy in the language comprehension task (a binary variable), and baseline-corrected pupil size. Fixed effects were age group (younger vs. older adults), surprisal (low vs. high surprisal), and task condition (single vs. dual). All predictors were sum coded.¹ We constructed separate linear mixed effects models for each dependent variable as implemented in the *lme4* library (Bates, Maechler, Bolker, & Walker, 2013; Version 1.1–15) in R (R Development Core Team, 2016; Version 3.4.3). Each model included random intercepts for participants and items. To reduce the risk of potentially inflated Type I error from an underspecified random slope structure, we fit each model using the maximal random slope structure warranted by the design (Barr, Levy, Scheepers, & Tily, 2013). In the case of nonconverging models, we simplified the random slope structure progressively using the *least variance* approach until convergence was achieved (see Barr et al., 2013; for guidelines). Higher-order interactions were followed up with planned model splits between younger and older adults. *p* values for each model were approximated using the Satterthwaite degrees of freedom method, as implemented in the *lmerTest* package in R. For table displays of model outputs, we used the *lmer* notation with vertical slashes to refer to random intercepts (e.g., by-subject random intercept: 1 | Subject) and random slopes (e.g., by-subject random slope for surprisal: Surprisal | Subject).

Tracking Task: Steering Deviations

There were two models to analyze steering deviations in the tracking task. The first model took into account tracking differences between single and dual-task conditions, not taking into account effects of surprisal (since surprisal is not specified for single-tracking blocks). The second model investigated tracking performance in the dual-task condition, specifically taking into account any potential effects of surprisal. Prior to analysis, steering deviations were trimmed for outliers at 3 *SD* above and below each individual's mean in the single- and dual-task conditions.

The model that compared single versus dual tracking (irrespective of surprisal) contained fixed effects for age group and task

Table 3

Effect Sizes (B), Standard Errors (SE), and t/p Values for Steering Deviations in the Single Versus Dual Tracking LMER Model

Fixed effects	Steering deviations			
	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
(Intercept)	.51	.06	7.97	‡
Task	.16	.05	3.23	**
Age group	.28	.13	2.19	*
Task × Age group	.19	.10	1.92	†
Random effects				Variance
1 Subject				.25
Task Subject				.16
Residual				1.16

† *p* < .1. ‡ *p* < 1. * *p* < .05. ** *p* < .01.

condition (see Table 3, for model output). We also included random intercepts for subjects; note that by-item random intercepts were not warranted for this model because there were no linguistic items in the single-tracking condition. The final model also included a by-subject random slope for task condition (i.e., the only within-subject predictor in the model).

There was a main effect of task condition (see Table 3), indicating that, regardless of age group, steering deviations were higher (time on target during tracking declined) during the dual versus the single tracking condition (see Figure 2). There was also a significant main effect of age group, indicating higher steering deviations in the group of older adults overall. The interaction between age group and task condition did not reach statistical significance in this model.

The model that investigated effects of surprisal specifically in the dual-tracking condition contained fixed effects for age group and surprisal. We also included random intercepts for subjects and items, by-subject random slopes for surprisal, and by-item random slopes for surprisal and age group. Table 4 summarizes the model output for the dual-tracking model.

There was a significant interaction between surprisal and age group (see Table 4). An inspection of Figure 3 suggests that this interaction was primarily driven by older adults, who demonstrated worse tracking for high-surprisal items. Steering deviations for younger adults, in contrast, did not seem to differ between low- and high-surprisal items. Indeed, these observations were borne out by a model split in which we split items by surprisal. A significant main effect of age group only emerged in the model for high-surprisal items ($b = 0.48$, $SE = 0.19$, $t = 2.47$, $p < .05$), and not in the model for the low-surprisal items ($b = 0.33$, $SE = 0.18$, $t = 1.88$, $p > .05$).

In sum, analysis of the steering deviations suggested two primary findings of interest. First, when comparing single versus dual

¹ In sum coding, the contrasts between levels of a two-level factor are represented as $-.5$ s and $.5$ s (as opposed to a dummy-, or treatment-coded model, where contrasts are represented by 0 s and 1 s). Therefore, the intercept in a sum-coded model represents the grand mean (not the baseline category as in a dummy-coded model), and factor coefficients can be interpreted as main, rather than simple, effects.

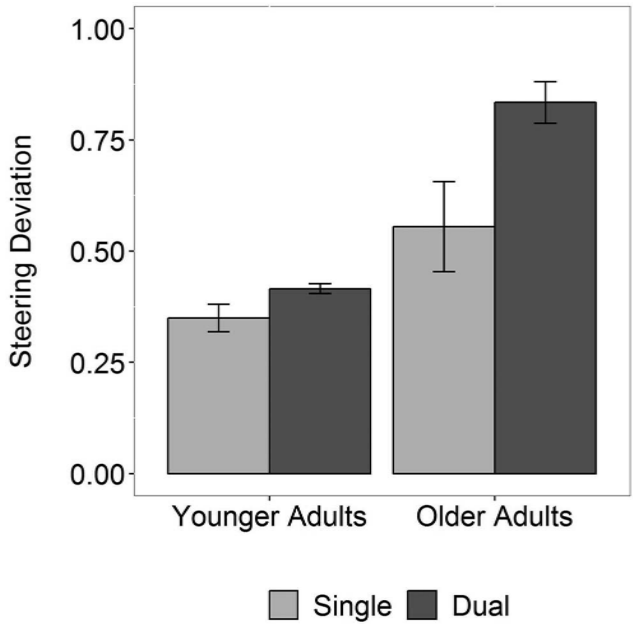


Figure 2. Steering deviations ($\pm SEM$) for single and dual tracking (irrespective of surprisal) in younger and older adults.

tracking, both younger and older adults showed an increase in steering deviations for the dual (tracking and simultaneous language comprehension) compared with the single task condition (tracking only), and steering deviations were greater for older adults in both conditions. Second, when comparing effects of surprisal specifically during the dual task, older adults showed an increase in steering deviations (decline in tracking) when processing high-surprisal (as opposed to low-surprisal) items.

Language Comprehension Task: Response Accuracy

In the model for response accuracy, the three-way interaction between surprisal, task condition, and age group did not reach

Table 4
Effect Sizes (*B*), Standard Errors (*SE*), and *t/p* Values for Steering Deviations in the Dual Tracking LMER Model

Fixed effects	Steering deviations			
	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
(Intercept)	.59	.09	6.68	‡
Surprisal	.10	.05	1.94	†
Age group	.40	.18	2.21	*
Surprisal × Age group	.19	.06	2.91	**
Random effects				Variance
1 Subject				.46
Surprisal Subject				.03
1 Item				.01
Surprisal Item				.09
Age group Item				.10
Residual				1.70

† $p < .1$. ‡ $p < 1$. * $p < .05$. ** $p < .01$.

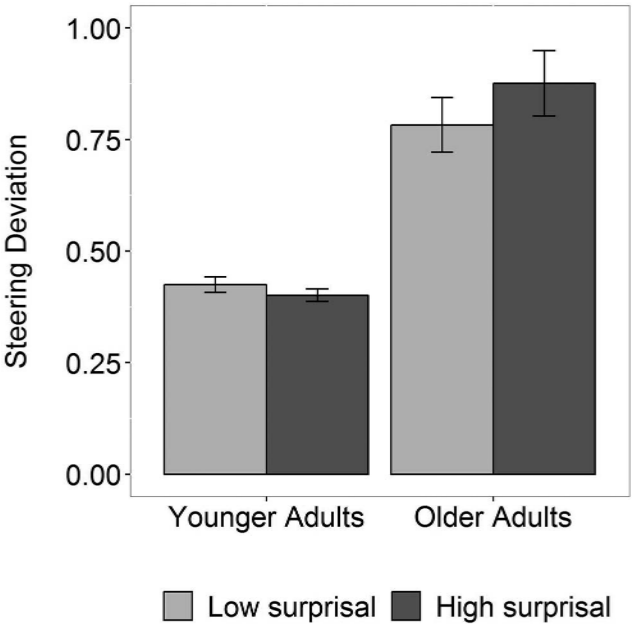


Figure 3. Average steering deviations in the dual task ($\pm SEM$), depending on age group and surprisal.

statistical significance, so it was excluded from the fixed-effects structure, $\chi^2(1) = 0.47$, $p = .50$. The final model for response accuracy contained random intercepts for participants and items; convergence issues with the binary model allowed for no random slopes. Table 5 summarizes the model output for the response accuracy *glmer* model.

We found a significant interaction between surprisal and age group (see Table 5). An inspection of the plot for this interaction (see Figure 4) suggested that younger adults showed lower accuracy for high-surprisal items overall (irrespective of task condition), whereas older adults demonstrated increased or relatively stable response accuracy for low- and high-surprisal items. This observation was borne out by another model split in

Table 5
Effect Sizes (*B*), Standard Errors (*SE*), and *z/p* Values for the Response Accuracy GLMER Model

Fixed effects	Response accuracy			
	<i>b</i>	<i>SE</i>	<i>z</i>	<i>p</i>
(Intercept)	1.58	.11	13.74	‡
Surprisal	-.22	.10	-2.12	*
Task	.09	.10	.91	‡
Age group	-.45	.16	-2.87	**
Surprisal × Task	.44	.23	1.90	†
Surprisal × Age group	.94	.21	4.54	***
Task × Age group	-.19	.21	-.93	‡
Random effects				Variance
1 Subject				.20
1 Item				.32

† $p < .1$. ‡ $p < 1$. * $p < .05$. ** $p < .01$. *** $p < .001$.

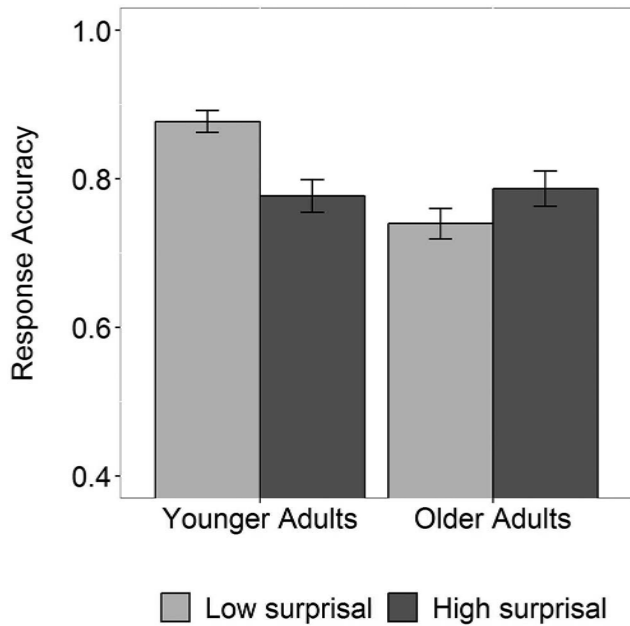


Figure 4. Average response accuracy ($\pm$ SEM) in younger and older adults for low- and high-surprisal items, averaged over task condition.

which we split subjects by age group: Only younger adults showed a statistically significant main effect of surprisal ($b = -0.68$, $SE = 0.15$, $z = -4.46$, $p < .001$), whereas older adults did not ($b = 0.27$, $SE = 0.14$, $z = 1.91$, $p > .05$).

In sum, the models for response accuracy showed that younger adults demonstrated lower response accuracy for high-surprisal items, irrespective of whether items were presented during the single- or dual-task condition. Older adults' response accuracy, in contrast, was similar for low- and high-surprisal items, irrespective of task demands.

Pupil Size

Eye-tracking data in both age groups were preprocessed by excluding missing data points (blinks, tracker loss) as well as gaze positions that fell outside of the screen. Raw pupil-size values were normalized (Piquado et al., 2010; Van Gerven et al., 2004) by subtracting the average pupil size per participant from the raw pupil values for that participant, and dividing the resulting value by 2.5 times the subject-wise standard deviation. Finally, we excluded trials for which data clean-up had resulted in a loss of more than 25%. Altogether, this resulted in a loss of 12% and 17% from the younger and older adults' data, respectively.

Based on visual inspection of the data and prior research from our lab (Demberg & Sayeed, 2016), we chose a time interval 700–1600 ms after critical word onset to analyze the pupil-size data statistically. Prior to analysis, pupil-size data were baseline-corrected by subtracting the average pupil values during the baseline region of 1000 ms before critical word onset (a common time interval; see Just & Carpenter, 1993; Piquado et al., 2010; Van Gerven et al., 2004) from the average pupil-size values during the peak region of 700–1600 ms after critical word onset. Statistical analysis was then performed on the resulting baseline-corrected scores. An overtime plot of the baseline-corrected pupil size-data is displayed in Figure 5.

The model for baseline-corrected pupil size in the critical time window (700–1600 ms after critical word onset) contained fixed effects for surprisal, age group, and task condition, as well as random intercepts for subjects and items. We also included by-subject random slopes for surprisal and task, and by-item random slopes for surprisal, task, and age group, as well as a nested random effect for eye within participants. Table 6 summarizes the model output for the pupil size LMER model.

There was a significant three-way interaction between surprisal, age group, and task condition (see Table 6). To explore the source of this interaction, we computed follow-up models in which we split subjects by age group. The model for younger adults showed a significant main effect of surprisal ($b = 0.06$, $SE = 0.02$, $t = 4.01$, $p < .001$), indicating larger pupil sizes for high-surprisal (as

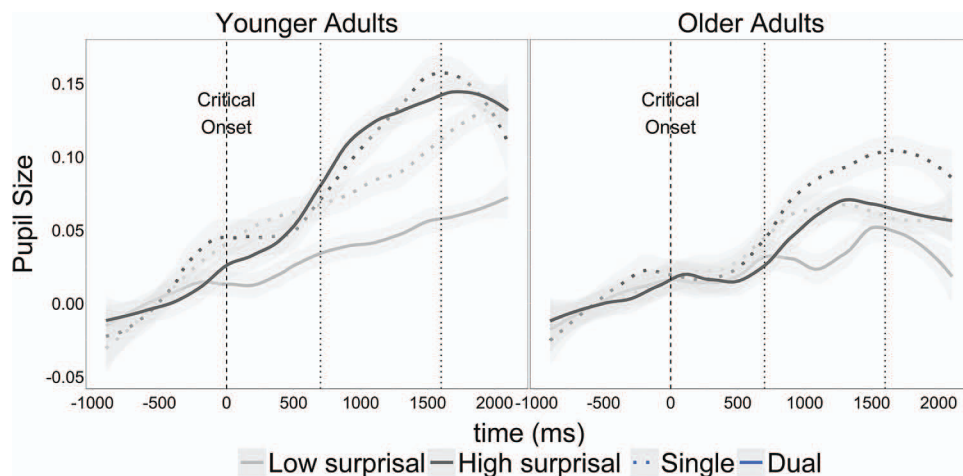


Figure 5. Baseline-corrected pupil size ($\pm$ 90% confidence bands), depending on age group and surprisal. Vertical dotted lines indicate the critical time window for statistical analysis.

Table 6
Effect Sizes (*B*), Standard Errors (*SE*), and *t/p* Values for the Pupil Size (Baseline-Corrected) LMER Model

Fixed effects	Baseline-corrected pupil size			
	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
(Intercept)	.08	.01	10.18	‡
Surprisal	.04	.01	4.11	***
Task	−.03	.01	−2.85	**
Age group	−.03	.01	−2.29	*
Surprisal × Task	−.02	.01	−2.12	*
Surprisal × Age group	−.03	.01	−2.45	*
Task × Age group	.00	.01	−.26	‡
Surprisal × Task × Age group	−.07	.02	−4.86	***
Random effects				Variance
1 Subject × Eye				.004
Surprisal Subject × Eye				.004
Task Subject × Eye				.005
1 Item				.001
Surprisal Item				.003
Task Item				.004
Age group Item				.003
Residual				.045

‡ $p < .1$. * $p < .05$. ** $p < .01$. *** $p < .001$.

opposed to low-surprisal) items in the younger adults group overall. The main effect of task condition did not quite reach statistical significance in this model ($b = 0.03$, $SE = 0.02$, $t = -1.90$, $p = .06$).

The model for the older adults, in turn, demonstrated a significant interaction between surprisal and task condition ($b = -0.06$, $SE = 0.01$, $t = -4.64$, $p < .001$). Another split of this model, in which we split items by task condition, demonstrated a significant main effect of surprisal only during the single-task condition ($b = 0.05$, $SE = 0.02$, $t = 2.67$, $p < .01$), but not during the dual-task condition ($b = 0.02$, $SE = 0.02$, $t = 0.92$, $p > .1$).

Thus, the models for baseline-corrected pupil size in the critical time window showed two primary findings of interest. First, pupil sizes in younger adults indicated greater cognitive load during processing of high-surprisal rather than low-surprisal items, regardless of whether participants were additionally performing a secondary task. Second, pupil sizes of older adults indicated higher processing load for high-surprisal items only when performing the language task in isolation, and not when additionally performing the tracking task.

Because the baseline-correction of the pupil-size data seemed to eliminate all effects of task condition in younger adults (most likely because task effects on the pupillary response were pervasive throughout the sentence and unrelated to critical word onset) we computed a follow-up model on the normalized (i.e., not baseline-corrected) pupil size data by averaging over surprisal, just taking into account effects of task condition and age group (for a bar-graph of these data, see Figure 6). This model contained by-subject random intercepts, as well as by-subject random slopes for task condition. By-item random intercepts and slopes were not considered because surprisal was of no interest in this model. There was nothing but a main effect of task condition (see Table 7, for model outputs), suggesting that irrespective of age group, pupil sizes were larger in the dual as opposed to the single task condition. No further effects reached significance in this model.

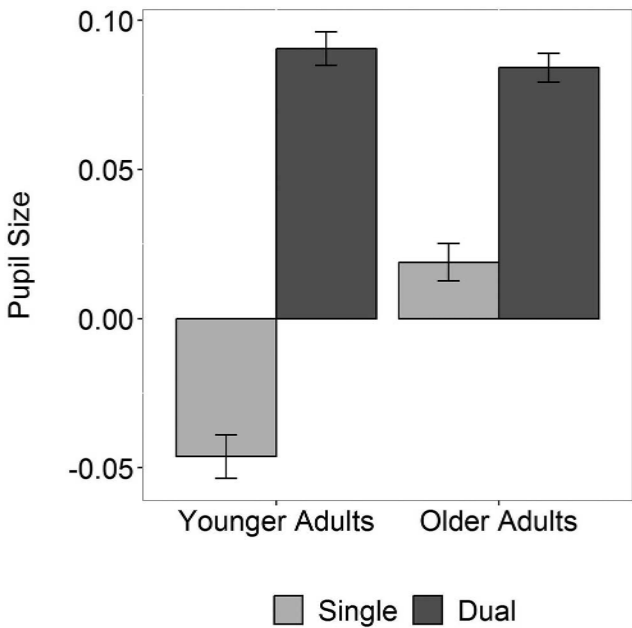


Figure 6. Normalized pupil size ($\pm SEM$) in younger and older adults, depending on task condition.

Discussion

Successful comprehension of spoken language is resource-consuming and dependent on a range of speech-inherent and environmental factors, for example the informational density of the speech signal (e.g., how surprising a word is, given its context), and/or whether language is comprehended in isolation or during the simultaneous execution of other tasks. Here, we investigated how surprisal and dual-task demands jointly affect language comprehension in older adults, a group that is inherently less able to perform multiple tasks at the same time (Verhaeghen & Cerella, 2002), including tasks that involve language (e.g., Becic et al., 2010; Kemper et al., 2003, 2009, 2010, 2014). In high-surprisal sentences, a sentence context strongly biased the expectation of a

Table 7
Effect Sizes (*B*), Standard Errors (*SE*), and *t/p* Values in the LMER Model for Normalized Pupil Size to Account for Effects of Task Condition

Fixed effects	Normalized pupil size			
	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
(Intercept)	.04	.01	8.14	‡
Task	.10	.02	5.26	***
Age Group	.02	.01	1.45	‡
Task: Age group	−.06	.04	−1.39	‡
Random effects				Variance
1 Subject				.00
Task Subject				.04
Residual				.08

‡ $p < .1$. *** $p < .001$.

particular target, but was completed with a semantically implausible target that violated contextual expectancies, thus introducing integration difficulties and cognitive load (Hale, 2001; Levy, 2008). To tap into the time course of online sentence processing, we used pupillometry to index rapid changes in pupil size as sentences unfolded acoustically (Beatty, 1982; Beatty & Lucero-Wagoner, 2000). We also measured participants' offline response accuracy in meaningfulness judgments (elicited from subjects after each sentence had ended), as well as their performance in the secondary task that required continuous tracking of a target. Our expectation was that the joint demands of integration difficulties associated with high-surprisal items (Jaeger, 2010; Levy & Jaeger, 2007) on the one hand, and the dual-task demands of splitting attention between two tasks on the other hand, would differentially affect older adults, by causing performance decline and possibly trade-offs among tasks in this participant group. As we describe below, our findings confirm these hypotheses, suggesting that that aging impairs sentence comprehension when integration difficulty (surprisal) and demands on attentional resources are high.

Consider first the task performance in younger adults. As a group, younger adults demonstrated strong dual-task costs, indicated by an increase in steering deviations (deviations from tracking target) during the dual-task (as opposed to the single-task) condition. Younger adults also showed effects of surprisal, because they responded less accurately to high-surprisal items across the board, suggesting that the higher informational density conveyed by these sentences indeed led to integration difficulties. Interestingly, however, younger adults showed no interaction between surprisal and task condition in either their tracking performance or in their response accuracy. This suggests that younger adults, despite the additional dual-task demands, had sufficient cognitive capacities to resolve the integration difficulty present in high-surprisal items—potentially because there was little processing overlap between input and output modalities in the dual task (i.e., auditory-visual input vs. motor output; see Levy & Pashler, 2001).

The physiological data obtained for younger adults largely confirm this behavioral pattern. First, the pupillary response in younger adults was greater for high-surprisal than low-surprisal items during both single- and dual-task conditions (with a peak at around 700–1600 ms after critical word onset), attesting to the greater comprehension difficulty associated with resolving these high-surprisal items. Second, there were strong effects of task condition in the younger adults' pupil size data (indicated by the main effect of task condition on the normalized pupil size values), with more elevated levels of pupil size during dual-task performance. In sum, younger adults as a group seemed to be able to maintain language comprehension and tracking without trade-offs between the tasks, maybe because of the lack of interference in input and output modality in our experiment (e.g., Hazeltine, Ruthruff, & Remington, 2006).

Compare this to the performance of the older adults, where processing of high-surprisal items was differently affected by single- versus dual-task demands in both behavioral and pupil size data. On a behavioral level, older adults showed a decline in tracking performance (i.e., an increase in steering deviations) when comprehending high-surprisal items during the dual task, whereas their response accuracy remained constant for both low- and high-surprisal items, irrespective of task condition. On a physiological level, older adults showed effects of surprisal (elevated

pupil size in online processing of high-surprisal sentences) only during the less demanding single-task condition, whereas during the dual-task condition, the pupillary response in older adults was similar low for both high- and low-surprisal items.

Taken together, these findings (i.e., flat pupillary response in older adults during online processing of high-surprisal sentences, but constant response accuracy at later stages of sentence comprehension, at the expense of tracking) indicate that older adults did not thoroughly process high-surprisal items during early, online stages of sentence processing, and then “overcompensated” for this initial failure at later, integrational stages of the task, by neglecting the tracking task while maintaining response accuracy in resolving the semantic mismatch in high-surprisal items. Thus, the joint effects of language that is higher in informational density and a dual-task situation seemed to exceed attentional resources in older adults, resulting in trade-offs between early and late stages of language processing, and also between language comprehension and tracking.

Age-related trade-offs between early and late stages of language processing have frequently been documented for language that is higher in informational density. For example, Stine and Hindman (1994) used a sentence-reading paradigm and subsequent recall task, and found that older adults, when presented with sentences high in propositional density (operationalized as the number of idea units in a sentence), took more time than younger adults in reading and encoding stimuli upon first presentation, but showed young-like performance in later recall of the same sentences. Thus, older adults allocated more time to encoding the sentences first-pass, and then benefitted from that extra time in organizing their idea units for subsequent recall in a way very similar to younger adults, an effect that might be compensatory in nature (cf. Payne & Stine-Morrow, 2012; Payne et al., 2014).

In an experimental set-up like ours, where the auditory input signal is fleeting and no extra time can be spent on encoding stimuli online, attentional resources can only be allocated once the auditory signal has ended, possibly at the later stage of comprehension and semantic integration when subjects are asked to judge the meaningfulness of the sentence. Indeed, our data suggest that it was precisely during those later stages of sentence comprehension when time was more freely available that older adults processed high-surprisal items more thoroughly, by allocating most attentional resources onto sentence comprehension while neglecting tracking. Younger adults, in contrast, seemed to register the semantic anomaly in high-surprisal items during much earlier stages of language processing, because their pupillary response suggested cognitive load and comprehension difficulty already at 700 ms after critical word onset (i.e., already while the sentence was playing).

Thus, the joint use of online and offline measures in the present study allows for insights into (a) age-related differences in the time course with which semantic information becomes available and is integrated during more demanding task situations and (b) age-related differences in the adjustment to dual-task demands: Whereas younger adults register and integrate semantic violation right away, older adults process them later and then overengage in repairing them (cf. Shake & Stine-Morrow, 2011, and Stites et al., 2013, for converging results on expectancy violations in anaphor and ambiguity resolution, respectively), at the expense of the secondary task. Thus, our data are in line with a large body of research suggesting that older adults self-regulate language pro-

cessing differently than younger adults (cf. Stine-Morrow et al., 2006, 2008), and that older adults show different patterns of task prioritization and neglect than younger adults, by trading off one task against another to maintain performance (e.g., Becic et al., 2010; Kemper et al., 2009; Li et al., 2001).

The Nature of Age-Related Decline in Processing Linguistic Load

So what is it about high-surprisal language that exceeds the capacity of older adults to such an extent? Stine-Morrow et al. (2008) describe language processing as a process that makes connections between concepts or idea units. This process is facilitated when contextual support provides a schema for connecting new incoming information, but it is prolonged and effortful when there is no contextual support, or when context and incoming information mismatch and induce integration difficulty, as in the current experiment. The fact that older adults are disproportionately affected by language-induced processing difficulties in a dual-task paradigm may be related to age-related declines in fluid aspects of cognition (Braver & Barch, 2002; Braver, Paxton, Locke, & Barch, 2009; Craik & Salthouse, 2000; Hasher, Zacks, & May, 1998) that render older adults less able to quickly integrate semantically conflicting information when attentional demands are already high. Indeed, cognitive impairments in older adults have been attributed to a range of age-related differences in the allocation of attentional resources during language processing (e.g., Stine-Morrow et al., 2008). For example, during reading, younger adults commonly show wrap-up effects (lingering or dwelling, suggesting conceptual integration) at major sentence or clause boundaries, whereas older adults pause more frequently and at smaller text intervals, thereby “chunking” conceptual information into more piece-meal amounts and effectively reducing cognitive load during semantic integration (Stine-Morrow & Miller, 2009). In addition to such age-related impairments in fluid aspects of cognition, a large body of research has shown that older adults rely more on top-down, context-based processing (Miller & Stine-Morrow, 1998), maybe as a compensatory strategy for their fluid impairments and sensory declines in bottom-up processing (hearing or vision impairments; cf. Payne & Stine-Morrow, 2012). Thus, high-surprisal sentences may be especially problematic for older adults because of older adults’ greater sensitivity to message- and text-level conceptual representations, as well as age-related fluid impairments in quickly integrating conflicting information.

Context Effect or Integration Effect?

One question that remains unanswered by the present investigation is whether the late-stage surprisal effects in our group of older adults (constant response accuracy at the expense of tracking) were driven by an overall integration cost associated with processing semantically conflicting information, or a cost related to violations of contextual predictions in the first clause of the sentence. In other words, we cannot dissociate whether older adults, when facing the dual-task demands, started to heavily rely on contextual information to predict upcoming information, resulting in tracking decline when high-surprisal words violated contextual expectations, or whether the observed surprisal effects were just a processing cost for integrating semantically conflicting in-

formation, independent of the contextual anticipation of specific words or concepts. Even though older adults’ well-documented increase in conceptual, top-down would presumably suggest that our findings are driven by a context effect (in other words, greater predictive processing in older adults, followed by re-processing and repair when predictions were disconfirmed), there is also evidence in the literature (primarily during the early N400 time window from ERP studies) that older adults are globally less sensitive to contextual constraints than younger adults, and actually rely less on predictive processing (Federmeier et al., 2002; Wlotko et al., 2010, 2012). Thus, the question concerning the locus of the functional impairment in older adults remains somewhat open here.

Finally, it is interesting to note that the late-stage task trade-off in this study (language comprehension at the expense of tracking) was opposite in direction to the one reported in an earlier dual-task study (Becic et al., 2010), which also investigated age-differences in language processing and simultaneous driving. In contrast to our findings here, the older adults from that particular study were more likely to focus on the secondary driving task, while neglecting the language task. One possible explanation for these contrasting findings relates to differences in task design. The bar-tracking task in the present study (see Figure 1) may have been less realistic, because it involved no other cars or road signs, just a straight road and two vertical bars. In contrast, Becic et al. (2010) used a highly naturalistic driving scenario with commonly encountered traffic situations, for example lane changes, overtaking maneuvers, and stops at traffic lights. In hindsight of prior studies arguing that task trade-offs in older adults are largely modulated by “survival value” (Li et al., 2001; Lindenberger et al., 2000), the bar tracking task in our study may have seemed expendable when compared with a task that involved overt verbal responses—responses which could be right or wrong. Older adults may have focused primarily on the language and neglected the tracking task to avoid mistakes, which are more easily observable, and because of their higher subjective priority to “save face” in the presence of the tester who coded their responses.

Conclusion

In sum, our data show that processing words that are high in surprisal during a dual task (here, operationalized as a language comprehension and tracking task) leads to task trade-offs between language and tracking in older adults, possibly because the integration difficulties associated with high-surprisal items add to age-related impairments in splitting attention between two tasks, causing cognitive load. Our findings support the use of surprisal as a metric for integration difficulty and cognitive load, and add to a growing body of research suggesting that (a) higher informational density disproportionately affects older adults when demands on attentional resources are already high and (b) older adults self-regulate attention allocation during language processing by trading off early processing against late repair.

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